

Post-Quantum Cryptography Migration

Part 17 - Edges & Sectors: Financial Messaging & Sector Rails

Compiled by the Spaced Research Ledger

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The world's money moves through a small set of rails: SWIFT messages between banks, central-bank settlement systems, and the card networks. Their cryptography is old, their message formats are tight, and a successful quantum attack on any of them would be a macroeconomic event. One study put a theoretical attack on the U.S. settlement system at a 10 to 17 percent hit to GDP.

This report covers the rails in three sections. Section 1 covers SWIFT and the ISO 20022 messaging standard, Section 2 the card networks and settlement infrastructure, and Section 3 the standards bodies writing the sector's rules.

The year's landmark is a real test on a real system. Central banks ran post-quantum signatures through the Eurosystem's TARGET2 settlement platform, and everything worked, at a price. Signatures came out thirteen times larger than RSA, verification ran seven and a half times slower, and message buffers broke along the way. SWIFT itself is expected to be post-quantum-enabled by 2027.

The coordination machinery arrived with it. The U.S. Treasury launched a quantum-readiness task force for the sector in August, the G7 published its coordinated roadmap, and Europe's payments council rewrote its cryptographic guidelines around the transition.

1. SWIFT & ISO 20022

SWIFT carries the payment instructions between the world's banks, and ISO 20022 is the message format the industry has just finished adopting. Both were sized for classical signatures. This section covers the first live post-quantum tests in a settlement system, and the surprisingly tight arithmetic of fitting new signatures into old messages.

Current State

Project Leap made the migration empirical. The central-bank collaboration spanned the BIS innovation hub, the Bank of Italy, the Bank of France, the Bundesbank, and SWIFT [1]. It tested post-quantum cryptography in an operational payment system, publishing in December 2025 [1]. Phase 2 replaced traditional signatures on liquidity transfers in TARGET2, modifying numerous components for compatibility [2]. Every scenario succeeded, while revealing performance differences that demand more testing before a systemic transition [2].

The specifics are sobering: a Dilithium signature of 3,293 bytes replaced a 256-byte RSA signature in the message header, a thirteen-fold increase [3]. Testing used the pre-standard algorithm because the finalized ML-DSA could not yet be exercised [3]. The larger payloads overflowed TARGET2's expected buffers, requiring engineering rework [3], and verification took meaningfully longer, forcing capacity replanning for

high-frequency settlement [4]. Phase 1 had earlier moved test payments through a quantum-resistant VPN between Paris and Frankfurt [5].

The message-size arithmetic drives algorithm choice. A simulation study of correspondent banking concludes Falcon-512 is the only NIST signature fitting SWIFT's legacy 2,048-byte message limit [6]. It recommends ML-KEM for key establishment with the level set by risk appetite [6], and rules out SLH-DSA for high-frequency signing on performance [6]. SWIFT's own path is dated: its next-generation platform is expected post-quantum-enabled by 2027, with a 15-month migration window for institutions [3].

The surrounding obligations are indirect but real. No explicit U.S. mandate requires financial post-quantum deployment, though risk-based security programs already cover cryptographic control [7]. Europe is more explicit, with the coordinated roadmap asking for transition starts by end-2026 and critical financial infrastructure secured by 2030 [8], reinforced by the operational-resilience and network-security regulations [8].

HSMs sit under all of it, payment and general-purpose alike, and must gain the algorithms because they anchor banking trust [9]. The X9 standards committee covers the messaging side, with a readiness assessment spanning both card and ISO 20022 formats [10]. Its key-management standard applies immediately to structured payment data [11], and a standing technical report covers quantum techniques in messaging [12].

2. Card Networks, Clearing & Settlement

Behind every card swipe sits a clearing and settlement stack that nets and moves the money. Its operators are infrastructure companies with decade-scale change cycles, now being convened by treasuries and central banks. This section covers who is coordinating, who has published, and the one network with a paper trail.

Recent Developments

The coordination bodies formed. The U.S. Treasury launched its Quantum-Readiness Task Force in August 2026, a public-private effort with workstreams on sector alignment, vendor readiness, and market-infrastructure risk [13]. It explicitly spans clearing houses and settlement operators [13].

A community migration framework split payments into its own extension document, acknowledging the complexity of cross-border flows and card dependencies [14]. On the standards side, the new QR-payment standard signs with the X9 financial PKI, positioned as a bridge to post-quantum signatures [15].

Current State

Mastercard is the network with public research. Its white paper with NTU Singapore concludes post-quantum cryptography beats quantum key distribution for the sector's harvest-now-decrypt-later problem, with cryptographic inventory the immediate priority [16]. It names agility the cornerstone of readiness, recommending algorithm-selection abstraction layers in payment infrastructure [3]. Visa maintains a research track with no public product announcements, and the other networks show nothing beyond EMVCo

participation [3]. The financial PKI those networks rely on is governed by the X9 committee through its certificate products [17].

The settlement layer has its stakes quantified. The Hudson Institute's estimate for a quantum attack on Fedwire runs to a possible six-month recession [18]. The BIS has published its quantum-readiness roadmap for the financial system [19], and Norway's central bank expects extensive implementation with entity responsibility plus coordination [20]. The TARGET2 numbers give the engineering reality: functional success with a 7.5-times verification slowdown and a need for parallel post-quantum certificate distribution across participants [3]. The G7's cyber experts published the sector's coordinated transition roadmap in January [21], followed by the first quantum-readiness report from its technologies working group [22].

The vendor and guidance layer is assembling. Quantum-safe HSMs are marketed across settlement, instant payments, and central-bank digital currency infrastructure [23], with validated hardware supporting the NIST algorithms [23] and a joint consulting-plus-hardware offering for payments [23]. Industry analysis puts the burden on identifying high-risk areas and prioritizing remediation across the market infrastructures [24], with agility embedded into settlement and digital-currency systems [18]. The X9 assessment covers both messaging formats used in clearing [17][3], though no dedicated engineering analysis of field-level impact on the card-clearing format yet exists [3]. An X9 working group is building assessment guidelines to identify which financial standards need updating [25].

3. Financial Standards Bodies

Three bodies write the rules the payments industry actually follows: X9 in the United States, EMVCo for chip cards, and the European Payments Council for the eurozone. Their positions range from urgent to explicitly unhurried. This section covers what each has published and where they differ.

Recent Developments

Europe's guidelines took their post-quantum turn. The payments council published version 16.0.1 of its cryptographic-algorithms guidance in June 2026 [26], following the March edition that significantly expanded its post-quantum content [27].

Current State

X9 has the longest institutional runway. It formed its quantum study group in 2017 [12] and published its technical report introducing quantum consequences for financial services in 2019 [28][29]. It has assigned work to extend its algorithm-agnostic messaging standard for the transition [30]. Its 2025 readiness assessment gives management guidance on identifying and prioritizing quantum risk [12][31], and its cryptographic-tools group is building assessment guidelines usable by auditors and regulators [25].

EMVCo is the deliberate outlier. Its published position expects no practical quantum threat to EMV before 2040, possibly never, while monitoring [32], through a public position statement and an engaged security working group [33]. The reasoning is structural: online transaction cryptograms are symmetric AES, present since 2010 and effectively future-proof, while legacy triple-DES retains a smaller margin [34]. The exposure it concedes is offline transactions, which use quantum-vulnerable RSA and elliptic curves [34].

The engagement is increasing anyway, with post-quantum in its chip workstream since 2023 [35]. A special interest meeting on quantum resilience ran in March 2026 [36], and two further papers are planned on market requirements and offline mitigations [37].

The European Payments Council writes the operational middle ground. Its annually reviewed guidelines reflect the cryptographic state of the art including post-quantum [38], reference the U.S. securities regulator's post-quantum roadmap and European symmetric-cryptography analysis [26], and stay scoped to algorithms and key management for interoperable cross-border payments [26]. Its earlier industry guidance captured the transitional consensus: hybrids to ease the path, elliptic curves as the immediate card upgrade [39].

About This Report

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